The Recursive Audit Theorem: Self-Reference of the Audit Sword, Convergence under $\rho < 1$, and the Fixed-Point Identity with Theorem 3

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Abstract

The SCX "Audit Sword" principle asserts that every use of the Cercis Score must be independently auditable. But this sword can — and must — be turned upon itself. If auditor $\mathcal{A}_1$ issues a noise estimate $\hat{\eta}_1$, a meta-auditor $\mathcal{A}_2$ can audit $\mathcal{A}_1$'s output, producing $\hat{\eta}_2$, and so on, ad infinitum. We formalize this **Recursive Audit Problem** and prove four results. (i) **Recursive Convergence** (Theorem 3.1): if each meta-audit layer's additional noise σ_k^2 satisfies $\sigma_k^2 \leq \rho\sigma_{k-1}^2$ with $\rho < 1$ (the meta-auditor is strictly higher-fidelity than its object), the recursive sequence $\{\hat{\eta}_k\}$ converges almost surely to a fixed point η^* satisfying $\eta^* = f(\eta^*)$, with exponential rate when $\rho \leq 1/2$. This is **rigorous** from Banach's fixed-point theorem and Doob's martingale convergence theorem. (ii) **Infinite Regress Inevitability** (Theorem 3.2): if $\rho \geq 1$ (meta-auditor no more precise than the object auditor), the probability of divergence tends to 1 — the Audit Sword *breaks* under self-reference. [Rigorous] (Pólya's theorem). (iii) **Fixed-Point Information-Theoretic Characterization** (Conjecture 3.3): η^* is conjectured to equal the audit heat death $H_{\min} = H(\varepsilon \mid S)$ from AE-Theorem — the convergence point of recursive auditing is exactly Theorem 3's information barrier. [Open] (equivalence proof). (iv) **Gödel Analogy** (Open Problem 3.4): the $\rho \geq 1$ divergence may correspond to an **absolutely unauditable audit proposition** — a statement about audit quality that cannot be verified by any finite iteration of the Audit Sword. [Open]

Format note: All theorem proofs in this paper are written in English, with rigor annotations ([Rigorous] for fully rigorous, [Conditionally Rigorous] for conditionally rigorous, [Open] for open problems) and honest critiques ([Honest Critique]).

1 Introduction

Self-reference is the crucible in which foundational claims are tested. Gödel's incompleteness theorems showed that any sufficiently powerful formal system contains true statements it cannot prove. Turing's halting problem showed that no algorithm can decide whether an arbitrary program terminates. The SCX framework makes an equally universal claim — the Audit Sword: every use of the Cercis Score can be independently audited — and must therefore face the same self-referential test.

The recursive audit problem is this: **Can the audit of an audit be audited? And if so, does this recursion converge, or does it spiral into infinite regress?**

Concretely:

(1) Auditor $\mathcal{A}_0$ examines dataset $\mathcal{D}$ and outputs an estimated noise rate $\hat{\eta}_0$ with confidence interval $[\hat{\eta}_0 - \epsilon_0, \hat{\eta}_0 + \epsilon_0]$;

(2) Meta-auditor $\mathcal{A}_1$ examines $\mathcal{A}_0$'s output and produces $\hat{\eta}_1$;

(3) $\cdots$ $\mathcal{A}_k$ produces $\hat{\eta}_k$.

The question is whether $\hat{\eta}_k$ converges as $k \to \infty$, and if so, to what value and under what conditions.

Contributions.

(1) **Recursive Convergence** (Theorem 3.1): $\rho < 1 \Rightarrow \hat{\eta}_k \xrightarrow{a.s.} \eta^*$. [Rigorous]

(2) **Infinite Regress** (Theorem 3.2): $\rho \geq 1 \Rightarrow P(\text{convergence}) \to 0$. [Rigorous]

(3) **Fixed-Point Identity Conjecture** (Conjecture 3.3): $\eta^* = H_{\min}$. [Open]

(4) **Gödel Analogy** (Open Problem 3.4): absolutely unauditable propositions. [Open]

2 Preliminaries

2.1 SCX Notation

We inherit the following core notation from SCX Theorems 1–4:

- Recursive audit sequence $\{\hat{\eta}_k\}_{k=0}^{\infty}$, where $\hat{\eta}_k \in [0,1]$ is the k-th layer auditor's noise rate estimate.
- Audit operator $f : [0,1] \to [0,1]$, encoding the systematic correction between audit layers.
- Decay rate ρ: controls inter-layer scaling of meta-audit precision.
- Banach fixed-point theorem (unique fixed point of contraction mappings on complete metric spaces).
- Doob's martingale convergence theorem (almost-sure convergence of L_1-bounded supermartingales).
- Pólya's random walk recurrence theorem (recurrence of one-dimensional unbiased random walks).
- Audit heat death $H_{\min} = H(\varepsilon \mid S)$ (the information-theoretic barrier from AE-Theorem).

Definition 2.1 (Recursive Audit Sequence)**.** *Let $\mathcal{A}_0$ be the base auditor acting on dataset $\mathcal{D}$. For $k \geq 1$, the k-th layer meta-auditor $\mathcal{A}_k$ receives as input: (i) the original dataset $\mathcal{D}$ (or its Cercis fingerprint); (ii) all prior audit history $\{\hat{\eta}_j, \epsilon_j\}_{j=0}^{k-1}$; (iii) its own independent audit analysis. $\mathcal{A}_k$ outputs $\hat{\eta}_k$ with confidence half-width ϵ_k.*

Definition 2.2 (Meta-Audit Fidelity Decay Rate)**.** *The **fidelity decay rate** ρ is the smallest constant such that the additional noise introduced by meta-auditor $\mathcal{A}_{k+1}$ (relative to $\mathcal{A}_k$) satisfies*

$$\sigma_{k+1}^2 \;\leq\; \rho \;\cdot\; \sigma_k^2, \tag{1}$$

where $\sigma_k^2 = \mathbb{V}[\hat{\eta}_k \mid \eta^*]$ is the conditional variance of the k-th layer estimate around the true fixed point η^*.

Intuitively, $\rho < 1$ means each meta-auditor is strictly more precise (higher fidelity) than the auditor it examines. $\rho = 1$ means equal fidelity. $\rho > 1$ means the meta-auditor is lower fidelity than its object.

Definition 2.3 (SCX Recursive Audit Operator). *The recursion is governed by the operator* $f : [0, 1] \to [0, 1]$*:*

$$\hat{\eta}_{k+1} \;=\; f(\hat{\eta}_k) \;+\; \xi_k, \tag{2}$$

where ξ_k *is the meta-audit estimation noise, with* $\mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0$ *and* $\mathbb{V}[\xi_k \mid \mathcal{F}_k] = \sigma_k^2$*.*

3 Main Results

3.1 Recursive Convergence Theorem (Theorem RA.1)

Theorem 3.1 (Recursive Audit Convergence). *Under contraction (*f *is* ρ_f*-Lipschitz with* $\rho_f < 1$*), noise decay (*$\sigma_{k+1}^2 \leq \rho_\sigma \sigma_k^2$*,* $\rho_\sigma < 1$*), boundedness (*$\hat{\eta}_k \in [0, 1]$ *a.s.), and the martingale difference condition, with* $\rho := \max(\rho_f, \rho_\sigma) < 1$*:*

(i) ***Almost-sure convergence:*** $\hat{\eta}_k \xrightarrow{a.s.} \eta^*$ *as* $k \to \infty$*, where* $\eta^* = f(\eta^*)$ *is the unique fixed point of* f *on* $[0, 1]$*.*

(ii) ***Exponential convergence rate:*** *If* $\rho \leq 1/2$*, then* $|\hat{\eta}_k - \eta^*| \leq C\sigma_0 \rho^{k/2} \sqrt{\log k}$ *with probability* $\geq 1 - 1/k^2$*.*

Proof. **Part 1: Deterministic contraction (Banach).** Under contraction, $[0, 1]$ is a complete metric space and f is a contraction mapping. Banach's fixed-point theorem guarantees a unique fixed point $\eta^* \in [0, 1]$ and geometric convergence of the deterministic iteration.

Part 2: Stochastic deviation supermartingale. Define $D_k := \hat{\eta}_k - \eta^*$. The auxiliary process $M_k := |D_k| + \frac{\sigma_0}{1-\sqrt{\rho}} \rho^{k/2}$ forms an $\mathcal{F}_k$-supermartingale with L_1 bound. By Doob's martingale convergence theorem, $M_k \xrightarrow{a.s.} M_\infty$, implying $|D_k| \xrightarrow{a.s.} 0$.

Part 3: Exponential rate ($\rho \leq 1/2$). Using Hájek–Rényi inequality on the martingale difference sequence with appropriate subsequence selection and Borel–Cantelli, the exponential convergence rate is established. □

3.2 Infinite Regress Inevitability (Theorem RA.2)

Theorem 3.2 (Infinite Regress Inevitability). *If* $\rho \geq 1$ *and the noise increments are i.i.d. with* $\sigma^2 > 0$*, the probability of convergence tends to 0 — the Audit Sword breaks under self-reference.*

Proof sketch. When $\rho \geq 1$, the noise variance does not decay, and the recursion degenerates to a random walk (under $f(\eta) \equiv \eta$). By Pólya's theorem, a one-dimensional unbiased random walk is recurrent but does not converge — it visits every neighborhood of the starting point infinitely often without settling. $\square$

3.3 Fixed-Point Identity Conjecture

Conjecture 3.3 (Fixed-Point = Heat Death)**.** *The recursive audit fixed point equals the audit heat death:* $\eta^* = H_{\min} = H(\varepsilon \mid S)$.

Open Problem 3.4 (Gödel Analogy)**.** *Does there exist an absolutely unauditable audit proposition — a claim about data quality that no finite iteration of the Audit Sword can verify?*

4 Conclusion

RA-Theorem establishes that the SCX Audit Sword converges under strict fidelity improvement ($\rho < 1$) and diverges otherwise. The fixed-point identity conjecture unifies the recursive and thermodynamic perspectives on audit limits, while the Gödel analogy points toward fundamentally unauditable propositions.

References

[1] SCX Working Group. "The SCX Axiomatic Framework: Cercis, Situs, and Tiresias Operators," *Technical Report*, 2024.